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University of Manouba

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Specifications Document

for the design and development project

ID: 200 Version: 01

Topic:

Any-Quantile Probabilistic Forecasting of Short-Term Electricity Demand

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1 Introduction

This project focuses on developing an advanced probabilistic forecasting system for electricity demand using state-of-the-art deep learning techniques. The system implements the any-quantile approach, enabling prediction of arbitrary quantiles with guaranteed coherence and reliability.

Key innovations include quantile-coherent N-BEATS architecture and comprehensive evaluation across probabilistic and point forecast metrics.The solution will be delivered as a web application with an efficient technology stack for real-time forecasting operations.

The system serves energy grid operators, traders requiring reliable uncertainty quantification for operational planning and risk management.

2 Requested Work

This section identifies the key stakeholders of the forecasting system and specifies both functional and non-functional requirements that guide the system design and implementation.

2.1 Actors

The system serves various stakeholders with distinct roles and responsibilities.These actors are categorized into primary and secondary groups based on their interaction frequency and criticality to core system operations.

2.1.1 Primary Actors

The following actors interact directly and frequently with the forecasting system:

1. **Energy Grid Operators**: Responsible for real-time grid management and load balancing operations.
2. **Energy Traders**: Professionals who make trading decisions based on electricity demand forecasts.

2.1.2 Secondary Actors

The following actors have less frequent but important interactions with the system:

1. **Energy Planners**: Strategic decision-makers for long-term infrastructure planning.
2. **System Administrators**: Technical staff responsible for infrastructure management and system maintenance.

Primary actors interact with the system daily for operational decision-making, while secondary actors use the system periodically for strategic planning and administrative tasks.

2.2 Functional Needs

The following table outlines the specific functional requirements for each actor category. These requirements define the core capabilities and features that the forecasting system must provide to support each user group's operational needs.

Actor	Functional Needs
PRIMARY ACTORS	
Energy Grid Operators	• Generate prediction intervals for 48-hour horizon • Trigger alerts when predicted demand exceeds capacity thresholds • Display historical forecasts vs. actual consumption • Provide real-time dashboard with automatic updates
Energy Traders	• Generate multiple quantile levels for risk assessment • Export forecast data • Allow custom quantile selection
SECONDARY ACTORS	
Energy Planners	• Generate long-term forecast scenarios with uncertainty • Access trend analysis and seasonality reports • View strategic planning dashboards • Export planning reports
System Administrators	• Manage user accounts and assign roles • Configure system settings • Access audit logs and security events
COMMON FUNCTIONAL NEEDS (ALL ACTORS)	
All Users	• Upload and manage multiple time series • Preprocess data automatically (handle missing values, outliers) • Authenticate securely and access role-based features

Table 1: Actors and their functional needs

2.3 Non-Functional Needs

Beyond functional capabilities, the system must meet critical non-functional requirements to ensure production-grade performance, reliability, and security.

2.3.1 Scalability Requirements

The system architecture must accommodate growing data volumes and user concurrency.

- Rescaling time series that don't have the same length or frequency.
- Concurrent user support for multiple simultaneous requests.

2.3.2 Reliability Requirements

The system must maintain high availability and robust error handling mechanisms.

- Fault tolerance with automatic recovery from training failures.
- Data integrity with consistent quantile predictions (monotonicity guaranteed).
- Error handling with alert trigger.

2.3.3 Security Requirements

The system must implement comprehensive security measures to protect sensitive energy data and ensure authorized access.

- Data protection with encryption of sensitive consumption data.
- Access control with role based authentication and authorization.

2.4 Domain Needs

The forecasting system must address specific requirements inherent to the energy and time series domains to ensure accuracy, reliability, and practical applicability.

2.4.1 Energy Domain Requirements

The system must account for the unique characteristics of electricity consumption patterns.

- Temporal patterns capturing daily, weekly, and seasonal consumption cycles.
- Exogenous variables integration (weather, calendar, event data).
- Peak management with special handling for demand peaks and anomalies.

2.4.2 Time Series Domain Requirements

The system must effectively handle the complexities of temporal data analysis and management.

- Multi-series management for handling heterogeneous time series.
- Trend and seasonality detection and processing.
- Managing timestamps across different countries.

3 Work Environments

This section describes the hardware and software infrastructure required for developing, training, and deploying the forecasting system.

3.1 Hardware Configuration

The project utilizes multiple computing resources to support development, training, and deployment activities.

3.1.1 Machine 1:

- **Processor**: Intel Core i7-10700K or AMD Ryzen 7 3700X (8 cores minimum)
- **Memory**: 16GB DDR4 minimum (32GB recommended)
- **Storage**: 500GB NVMe SSD for data and models
- **GPU**: NVIDIA RTX 3070 (8GB VRAM)
- **Network**: Stable internet connection for data downloads and updates
- **Training Infrastructure**:
  - Platform: Docker containerized environment
  - CUDA: Version 11.8+ for GPU acceleration
  - Shared Memory: 8GB (–shm-size=8g) for multi-process training

3.1.2 Machine 2:

- **Device Name**: OMENCVteam
- **Hardware Model**: HP OMEN by HP 40L Gaming Desktop GT21-1xxx
- **Processor**: 13th Gen Intel® Core™ i7-13700K (24 threads)
- **Memory**: 32.0 GiB
- **Graphics Card**: NVIDIA GeForce RTX 4070 Ti (PCIe / SSE2)
- **Disk Capacity**: 2.0 TB
- **Operating System**: Ubuntu 22.04.5 LTS (64-bit)
- **GNOME Version**: 42.9
- **Windowing System**: X11
- **Access Method**: TeamViewer remote connection

3.1.3 Virtual Machine (CCK):

- **vCPU**: 64 cores
- **RAM**: 256 GB
- **Disk**: 1 TB
- **vGPU**: NVIDIA A40-1Q (24 GB VRAM)

3.2 Software Environment

The development stack comprises modern programming languages, frameworks, and tools optimized for deep learning and web application development.

3.2.1 Programming Languages and Frameworks

The core technology stack includes the following languages and frameworks:

- **Python 3.8+**: Primary development language
- **PyTorch 2.0+**: Deep learning framework
- **PyTorch Lightning 1.8+**: High-level training framework
- **YAML**: Configuration management
- **LaTeX**: Documentation and report generation

3.2.2 Key Libraries and Tools

The project leverages essential libraries and development tools across multiple domains:

- **Scientific Computing**: NumPy, SciPy, Pandas
- **Time Series**: GluonTS, scikit-learn
- **Visualization**: Matplotlib, Seaborn, Plotly, TensorBoard
- **Configuration**: OmegaConf, Hydra
- **Containerization**: Docker, Docker Compose
- **Version Control**: Git, GitHub/GitLab
- **Development**: VS Code, Jupyter Notebook

3.2.3 Web Application Stack

The forecasting system will be delivered as a web application with the following technology stack:

- **Frontend**:
  - React 18+ with TypeScript for type safety
  - Next.js for server-side rendering and routing
  - TailwindCSS for modern styling
  - Plotly.js for interactive time series visualizations
- **Backend**:

  - Python-based backend

- **Database**:
  - PostgreSQL 15+ for relational data
  - Redis for caching and message queue

4 Provisional Chronogram of the Project

The project spans **14 weeks** from February 2026 to May 2026, structured in 7 distinct phases with overlapping activities for optimal resource utilization.

4.1 Gantt Chart

Phase	**February**				**March**				**April**				**May**	
Week	1	2	3	4	1	2	3	4	1	2	3	4	1	2
Literature Review	■	■												
Environment Setup & Data Prep		■	■											
Baseline N-BEATS Implementation				■	■									
Advanced Architecture Development						■	■	■						
Web Application Development								■	■	■				
Testing & Validation											■	■		
Documentation & Reporting													■	

Table 2: Project Timeline - February to May 2026